ET 8212 ELECTROMAGNETIC THEORY I

SUBMIT ON 15th MAY 2024

ASSIGNMENT I

QUESTION ONE

Determine the flux of $\mathbf{D} = \rho^2 \cos^2 \phi \, \mathbf{a}_\rho + z \sin \phi \, \mathbf{a}_\phi$ over the closed surface of the cylinder $0 \leq z \leq 1$, $\rho = 4$. Verify the divergence theorem for this case.

QUESTION TWO

For a vector field $\mathbf{A}$, show explicitly that $\nabla \cdot \nabla \times \mathbf{A} = 0$; that is, the divergence of the curl of any vector field is zero.

QUESTION THREE

Show that the vector field $\mathbf{A}$ is conservative if $\mathbf{A}$ possesses one of these two properties:

(a) The line integral of the tangential component of $\mathbf{A}$ along a path extending from a point P to a point Q is independent of the path.

(b) The line integral of the tangential component of $\mathbf{A}$ around any closed path is zero.

QUESTION FOUR

Derive the four Maxwell Equations:-

a) In point form
b) In integral form